Class 6

Biology

Olympiad Program
Volume - II

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6. Living Organisms - Characteristics & Habitat

Solutions

LEVEL I

1. Correct option (A)

Desert plants and animals are suited to dry conditions and vast temperature differences. Desert plants are adapted to their arid environment in many ways. Small leaves on desert plants help reduce moisture loss during photosynthesis. Small leaves mean less evaporative surface per leaf. In addition, a small leaf in the sun doesn't reach as high a temperature as a large leaf.

2. Correct option (A)

Acacia Nilotic is truly multipurpose nitrogen fixing leguminous tree in India commonly called as babul. Its timber is valued by rural folks, leaves and pod are used as fodder and gum has a number of uses.

3. Correct option (A)

Forests cover 31 percent of the world's land surface, just over 4 billion hectares. (One hectare = 2.47 acres.)

4. Correct option (D)

In Kurnool district we can see a bird called battameka pitta which flies over long distances to come here.

5. Correct option (D)

The main components of a habitat are shelter, water, food, and space. A habitat is said to have a suitable arrangement when it has the correct amount of all of these.

6. Correct option (C)

The methods of reproduction in fishes are varied, but most fishes lay a large number of small eggs, fertilized and scattered outside of the body. The eggs of pelagic fishes usually remain suspended in the open water. Many shore and freshwater fishes lay eggs on the bottom.

7. Correct option (A)

Avian migration is a natural process, whereby different birds fly over distances of hundreds and thousands of kilometres in order to find the best ecological conditions and habitats for feeding, breeding and raising their young.

8. Correct option (B)

Pistoia stratiotes is a free-floating, fast growing, obligate aquatic that can form vast, dense floating mats, covering the entire water surface of lakes and slow-moving rivers

9. Correct option (D)

Rivers are the arteries of the landscape that connect and sustain diverse freshwater wetlands including lakes, inland deltas, swamps and marshes. Safeguarding and restoring healthy, resilient freshwater systems are key to achieving global goals for food, water.

10. Correct option (A)

The lotus is an aquatic plant that grows in shallow fresh water, with flat round leaves that float on the surface and famously beautiful flowers. While the entire plant is edible, the root that grows underwater is particularly popular in Asia, and especially in Japanese cooking.

11. Correct option (B)

Clostridium perfringens, Staphylococcus, and Escherichia coli (E. coli; toxic strain). They inhibit health by triggering disease and promoting aging. Opportunistic bacteria cause no trouble when you are healthy, but have adverse effects upon the intestines when the body is weak.

12. Correct option (C)
Habitats are a very significant part of natural diversity. They offer species a living environment and provide people with essential ecosystem services.

13. Correct option (D)
The main components of a habitat are shelter, water, food, and space. A habitat is said to have a suitable arrangement when it has the correct amount of all of these.

14. Correct option (C)
The common pond skater can be seen 'skating' over the surface of ponds, lakes, ditches and slow-moving rivers. It is predatory, feeding on small insects by detecting vibrations in the water's surface. There are some species of beetles and dragon flies that may at some point in their life cycle live on the surface of water but maximum time they either hover or live on land surfaces.

15. Correct option (A)
The ocean is very, very deep; light can only penetrate so far below the surface of the ocean. As the light energy travels through the water, the molecules in the water scatter and absorb it. At great depths, light is so scattered that there is nothing left to detect.

16. Correct option (D)
Animals move from one place to another in search of food and to protect themselves from their enemies. they also move to escape from the harsh climate. To find shelter and to breed their young ones.

17. Correct option (A)
A cactus is able to survive in the desert due to the following features: (i) It has long roots that go deep inside the soil for absorbing water. (ii) Its leaves are in the form of spines to prevent water loss through transpiration. (iii) Its stem is covered with a thick waxy layer to retain water.

18. Correct option (A)
Grapevines require nutrients for healthy, sustained growth. Without adequate nutrition, vines may have weak growth and reduced yields. An inadequate nutrient supply may be associated with low (<5) or high (>7) soil pH, inadequate irrigation, or competition from weeds or cover crops.

19. Correct option (C)
Mussels are marine or freshwater molluscs that are primarily found in bottom water. They attach themselves to various surfaces such as rocks, shells, or other substrates on the seabed or riverbed. They live in this habitat by filtering water to obtain nutrients.

20. Correct option (A)
An earthworm is a soil-dwelling terrestrial invertebrate that belongs to the phylum Annelida.

21. Correct Matching is: a- Q;b- R;c- S;d- R;e- P
a) – Q) Vallisneria is a genus of freshwater aquatic plant, commonly called eelgrass, tape grass or valise.
b) – R) Terrestrial plants are the plants which grow on land. Mango tree, rose plant and apple tree are terrestrial plants.
c) – S) A mangrove is a shrub or tree that grows mainly in coastal saline or brackish water. Mangroves grow in an equatorial climate, typically along coastlines.
d) – R) cactus is able to survive in the desert due to the following features: (i) It has long roots that go deep inside the soil for absorbing water. (ii) Its leaves are in the form of spines to prevent water loss through transpiration. (iii) Its stem is covered with a thick waxy layer to retain water.
e) – P) Most monkeys live mainly on trees.

LEVEL – II

1. Correct option (C)
All living elements-the birds, animals and plants, forests, fisheries are biotic elements.

2. Correct option (B)
Broadly Biotic components are categorised into two categories: Autotrophs and Heterotrophs or Producers and Consumers. A third category decomposers also exists but it is a part of Heterotrophs. Producers are the plants in the ecosystem, which can generate their own energy requirement through photosynthesis, in the presence of sunlight and chlorophyll. Consumers include herbivores, carnivores, omnivores, Decomposers and Saprophytes. The herbivores are the living organisms that feed on plants.

3. Correct option (A)
The organisms which prepare their own food are called autotrophs or producers. They depend on a process known as photosynthesis for this.

4. Correct option (B)
Photosynthesis is a process by which green plants prepare their own food from carbon dioxide and water in the presence of sunlight and chlorophyll. The reaction involved is as follows: $6CO2(g)+6H2O(l) \rightarrow C6H12O6(s)+6O2(g)$.

5. Correct option (C)
Decomposers are heterotrophic microbes like bacteria and fungi that derive their nutrition by decomposing dead and decaying organisms.

6. Correct option (A)
All animals and fungi are heterotrophs as they can't produce their own food. They all depend on the autotrophs (primary producers) and on other organisms for food. Hence, also known as consumers.

7. Correct option (A)
Jaguars and Tigers are carnivorous mammals that feed mainly on fish (such as piranha), reptiles (like lizards or turtles), birds (turkeys), amphibians (like frogs), as well as small mammals like monkeys and deer.

8. Correct option (C)
As they only eat flesh and meat of other animals for their food, they are classified as carnivores.

9. Correct option (C)
Animals that can consume both plants and other animals for nutrition are called omnivorous animals. Examples: human beings, cockroaches, dogs, bears, cats.

10. Correct option (B)
Humans, bears, crows are referred to as omnivores.

11. Correct option (A)
Plants and animals are interdependent on each other. Plants depend on animals for pollination, and they absorb nutrients from decomposed animal bodies to grow. Animals depend on plants for food, shelter, protection, to get oxygen and so on.

12. Correct option (D)
Earthworms need their skin to breathe. Because their skin is gas permeable. Their skin is moist, and cutaneous respiration is the process of breathing via the skin.

13. Correct option (D)
In roots, stems & leaves the exchange of gases takes place through diffusion. The surface of young stem has stomata, while that of an older stem have lenticel for gaseous exchange. Stomata are mainly present on the surface of leaves apart from the epidermal

surface of young stem for gaseous exchange. In root exchange of gases occurs through root hair.

14. Correct option (A)

The habitats of the plants and animals that live on land are called habitat. Organisms living on the land are called terrestrial animals.

15. Correct option (A)

Terrestrial habitats are ones that are found on land, like forests, grasslands, deserts, shorelines, and wetlands. Terrestrial habitats also include manmade habitats, like farms, towns, and cities, and habitats that are under the earth, like caves and mines.

16. Correct option (A)

Desert sand is a good absorber and good radiator of heat. During the day it absorbs heat quickly and becomes very hot. At night sand lose all the absorbed heat quickly and become cold. Therefore, deserts are cold at night and hot during the day.

17. Correct option (C)

Transpiration is the process through which plants lose water through stomata. Stomata are usually present on the underside of the leaves. It is essentially evaporation of water from the leaves of the plants.

18. Correct option (B)

Succulent plants store water in fleshy leaves, stems or roots. All cacti are succulents, as are such non-cactus desert dwellers as agave, aloe, elephant trees, and many euphorbias. Several other adaptations are essential for the water storing habit to be effective.

19. Correct option (A)

Nocturnal animals are animals that are active during the night and sleep during the day.

20. Correct option (D)

camel have padded feet help camel to walk on hot and slippery sand without sinking.

21. Correct option (B)

High altitudes can cause low oxygen saturation levels or desaturation of an individual's blood. It happens because of low atmospheric pressure at high altitudes. Oxygen saturation levels refer to the extent haemoglobin is bound or saturated to oxygen.

22. Correct option (A)

Herbivores like horses, cows, and rabbits have rows of wide, flat teeth. This helps them in grinding and chewing leaves, grass, and other tough plant materials.

23. Correct option (A)

Many plants in the humid rainforest canopy are pointed, so that rain can run off the tips of the leaves. These “drip tips” keep the leaves dry and free of Mold.

24. Correct option (A)

Habitats whose mode of survival is in water are known as aquatic habitats. The habitats of plants and animals which lives in water are said to live in aquatic habitats. Ocean, ponds, lakes, and rivers are examples of aquatic habitats. If the water is freshwater, then it is called fresh water habitat.

25. Correct option (B).

Large air gaps exist in the parenchyma of aquatic plants to let them float on water. Such a parenchyma type is called aerenchyma. It is a type of tissue with air gaps. It gives hydrophytes buoyancy, allowing them to float to the water's upper surface. These air cavities of aquatic plants allow the plant to be light and buoyant which results in the easy floating of plants in the water.

7. Body Movements
Solutions

LEVEL-I

1. Correct option (D)

When muscles contract, they undergo several simultaneous changes, which is why the correct answer is D) All of these. First off, as muscles contract, the individual muscle fibers within the muscle bundle pull together, shortening the muscle. Movement is made possible by this shortening of the muscle fibers, which also produces force. Second, while contracting, muscles can stiffen up, especially if they are actively resisting an outside force. Increased muscle tone and the activation of motor units cause this stiffness. The compression of muscle fibers and increased tension within the muscle can also make muscles appear thicker during contraction. Therefore, when muscles contract, all of these changes take place, making it a complex physiological process necessary for many bodily functions, such as movement and posture maintenance.

2. Correct option (C)

The bony or cartilaginous structure in vertebrates that is attached to and supports the hind limbs or fins is known as the "pelvic girdle," as indicated in option C. In vertebrates, the pelvic girdle is an essential part of the skeletal system, and it differs in structure depending on the species. The pelvic girdle, which forms the hip bones in terrestrial vertebrates like mammals, including humans, is crucial for supporting the body's weight, serving as points of attachment for leg muscles, and allowing hind limb movement. Similar functions are performed by the pelvic girdle in aquatic vertebrates like fish, which stabilize and support the pelvic fins that facilitate swimming and movement. While options A (backbone), B (ribs), and D (vertebrae) are also important parts of the vertebrate skeletal system, they do not specifically refer to the structure supporting the hind limbs or fins, making option C the correct choice for this description.

3. Correct option (B)

The delicate, fleshy structures that surround and are found beneath the skin are known as muscles. They are in charge of posture, movement, and heat generation. Muscle fiber bundles form the structure of muscles. Long, thin cells that can contract and relax are called muscle fibers. A muscle that is attached to a bone shortens and pulls on that bone when it contracts. This is how muscles cause bones to move. The body cannot function without muscles. They enable us to move, keep our posture, and generate heat. We could not survive without our muscles.

Tendons, which connect muscles to bones through fibrous cords, are strong. Bands of fibrous tissue called ligaments attach bones to one another.

4. Correct option (D)

The answer, "All the above," accurately reflects that almost all our body movements depend on the combined function of muscles, bones, and joints. Joints enable the range of motion and flexibility needed for various types of movements, while bones provide structural support and act as attachment points for muscles. Muscles contract and relax to generate the force required for movement. Our ability to perform a wide range of activities, from the fundamental activities of daily life to more difficult physical tasks, is due to the complex coordination between our muscles, bones, and joints. Our body's musculoskeletal system is a marvel of functional integration, allowing us to effectively interact with and navigate our environment. Whether it's the contraction of our leg muscles during walking, the flexibility of our shoulder joints for reaching, or the stability provided by our spinal bones.

5. Correct option (C)

Cartilage, as identified in option C, is a type of connective tissue that can be considered a bone-like structure, but it possesses flexibility. Contrary to rigid bones, cartilage is malleable and offers some flexibility and shock absorption in various body parts. It acts as a structural element where it forms the joint surfaces, allowing for painless movement in places like the nose, ears, and ends of long bones. Although cartilage is not as hard as bone, the body still needs it to provide support and protection. Although neither of these structures is regarded as flexible like cartilage, ligaments (option A) are fibrous tissues that connect bones to other bones, and tendons (option B) connect muscles to bones. Therefore, option C, "Cartilage," which is a flexible tissue that resembles bone, is the appropriate response.

6. Correct option (A)

The skull is also called the "cranium," as correctly identified in option A. The bony cranium surrounds the brain in a protective framework, giving it vital support and preventing damage from the outside world. In addition to housing the brain, it also houses other sensory organs like the eyes and ears. It is made up of several bones that are tightly fused together to form a sturdy, protective enclosure for the brain. The largest portion of the brain is referred to as the "cerebrum" (option C), while the "brain" (option B) refers to the entire organ housed within the cranium. Therefore, when referring to the skeletal framework of the head, option A, "Cranium," is the appropriate response.

7. Correct option (B)

Muscle contraction causes the muscle fibers to shorten, causing the muscle to thicken and enlarge. The reason for this is that the muscle fibers are shortening as a result of pulling against one another. The muscle becomes thicker and more bowed the more the muscle fibers shorten.

The opposite of contraction is relaxation. The muscle fibers lengthen as a muscle relaxes, making the muscle thinner and less puffed up.

So, the muscle bulges due to contraction. Option A, relaxation, is incorrect because the muscle does not bulge when it relaxes. Option C, both A & B, is incorrect because only contraction causes the muscle to bulge. Option D, none of these, is incorrect because the muscle does bulge due to contraction.

8. Correct option (D)

Cartilage can be found in the vertebral column, nose, and ears. Hyaline cartilage, elastic cartilage, and fibrocartilage are the three types of cartilage found in the ear. The eardrum and the joints between the bones of the middle ear contain hyaline cartilage. The outer ear is composed of elastic cartilage. The joint between the jawbone and the skull contains fibrocartilage.

Both hyaline and elastic cartilage can be found in the nose. The wall that separates the nose into two nostrils, known as the septum, is made of hyaline cartilage. The pinna, which is the outer portion of the ear, and the nostrils both contain elastic cartilage.

Vertebrae, stacked one on top of the other, make up the vertebral column. The joints between the vertebrae are cushioned by intervertebral discs, which are made up of fibrocartilage.

9. Correct option (C)

These vertebrae are a part of the spine's vertebral column, which acts as the body's main structural support system. The cervical (neck), thoracic (upper back), lumbar (lower back), sacral (pelvic), and coccygeal (tailbone) regions are among the regions that make up the vertebral column. In a typical adult spine, which has 26 vertebrae in all, there are 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, and 4 fused coccygeal. These vertebrae offer

adaptability and stability, enabling a variety of movements and safeguarding the spinal cord, an essential component of the central nervous system.

10. Correct option (D)

The human ribcage plays a crucial role in protecting the heart and lungs, so the correct response is D) Both A & B. The sternum and ribs make up the bony structure known as the ribcage (breastbone). It creates a cage-like shield around the chest region. The ribs surround and protect the heart, a vital organ in charge of pumping blood throughout the body, as well as the lungs, which are essential for breathing. The ribcage is crucial in protecting the heart and lungs because any blow to the chest can potentially damage these important organs.

11. Correct option (B)

Vertebrae, not vertebrates, are the name for the tiny bones that run along the back and support the spinal cord. A group of animals known as vertebrates have a backbone or vertebral column made up of these individual vertebrae. In vertebrate animals, including humans, the spinal column is made up of bony segments called vertebrae. The delicate spinal cord, a vital component of the nervous system that transmits signals from the brain to the rest of the body, is shielded by these vertebrae, which also support the body's structural integrity. Therefore, the individual bones that make up the spinal column or backbone in vertebrate animals are referred to as "vertebrae."

12. Correct option (C)

Muscles and bones are the two main parts of our body that help in body movements.

The tissues that contract and relax to move the bones are called muscles. Tendons connect them to the bones. The human body has over 600 muscles, which account for about 40% of our body weight.

The rigid, brittle structures that give the body support and structure are called bones. Additionally, they shield the internal organs. The adult human skeleton has 206 bones.

In order to move, the muscles and bones must cooperate. The muscles supply the force to move the bones while the bones serve as a framework for the muscles to attach to.

Therefore, bones and muscles both play a role in how the body moves. They work together to produce the movements that we need to do everyday activities, such as walking, running, and eating.

13. Correct option (C)

The correct answer is C) Both A & B. Through a combination of expansion (stretching) and contraction, muscles are what move bones. When a muscle contracts, it shortens and pulls on the bone to which it is attached, moving that bone as a result. It is this contraction that produces the power required for various bodily movements. On the other hand, when a muscle contracts or shortens, it prevents the bone from moving or returning to its original position. As a result, muscles' expansion (relaxation) and contraction are both involved in the process of moving bones, making Option C the best option.

14. Correct option (A)

Ligaments are strong, fibrous connective tissues that provide a unique means of securing bones to one another. They are essential for stabilizing joints because they link one bone to another. Ligaments are crucial for preserving the skeletal system's structural integrity and ensuring that joints work properly. In contrast to muscles, which contract and produce movement, ligaments support joints and restrict excessive motion, thereby reducing the risk of dislocations and other joint injuries. Therefore, ligaments are in charge of securing bones so that joints remain stable and allow for coordinated, controlled movement.

The lower leg's calf bones are those. The fusion of bones is not their responsibility. Tendons connect muscles to bones. Collagen also makes up tendon tissue. The force of a muscle contraction is transferred to the bones by the tendons, which causes the bones to move.

15. Correct option (C)

The answer, "C) Pivotal," is accurate because the pivotal joint is specifically found in the neck region. This kind of joint permits rotational movement about a central axis, which is necessary for the neck's flexibility and mobility. It gives us the ability to turn our heads side to side and look in various directions. For many daily tasks and activities, such as driving, reading, talking, and taking in our surroundings, this rotational ability is essential. The pivotal joint in the neck is a crucial component of the musculoskeletal system of the human body because it allows for smooth and controlled rotation, enabling a variety of head movements.

16. Correct option (B)

Fixated joints, also referred to as sutures or immovable joints, are those where the bones are closely connected and do not permit any significant movement. These fixed joints or sutures hold the cranium's bones together in the human skull. This immobility is crucial for preserving the head's overall stability and shape while safeguarding the sensitive brain inside the skull. The joints in the vertebral column (Option A), the neck (Option C), and the shoulder (Option D) all allow for varying degrees of movement, whereas the joints in the skull (Option B), which are fixed, remain rigid to protect the brain and facial structures.

17. Correct option (A)

The correct answer is A) Forward push by pressing the ground. Snails have a distinctive and slow form of locomotion. They cover the underside of their bodies with a unique muscular structure called a foot. A snail moves by tensing the muscles in its foot, which causes it to push forward against the ground. Mucus produced by the foot reduces friction, allowing the snail to move forward more easily. Snails move slowly and steadily because of this forward and rhythmic pushing against the surface and the lubricating impact of the mucus. They don't move by jerking or forming food (Option B) (Option C). Therefore, snail movement is primarily accomplished by pressing their muscular foot into the ground.

18. Correct option (C)

A hinge joint is indeed found in the elbow and arms. Similar to how a hinged door opens and closes, this kind of joint allows movement primarily in a single plane. The elbow joint allows the forearm to bend and straighten in relation to the upper arm in the case of the elbow. It is essential for many daily activities, such as bending the elbow to lift objects or extending the arm to reach for something, that this hinge-like structure maintains stability while enabling controlled and relatively simple movement in a single direction. The hinge joint is thus appropriately named given its resemblance to a hinge and its function in allowing the elbow and arms to move.

19. Correct option (A)

Tendons are the connective tissues that join muscles to bones. They act as the connecting element that enables the force produced by contracting muscles to be transferred to the bones, resulting in movement at the joints. Tendons are strong and adaptable, built to withstand the stress and tension brought on by muscle contraction. Contrarily, cartilage (Option C) is a different type of connective tissue that covers the ends of bones at joints and provides cushioning and support, whereas ligaments (Option B) connect bone to bone and stabilize joints. So, Option A is the best choice because tendons are specifically responsible for joining muscles to bones.

20. Correct option (D)

The average infant's backbone contains 33 vertebrae. Considering that infants have extra, unfused vertebrae, this number is different from the 26 vertebrae that make up the adult spine. In contrast to adults, infants have 5 unfused sacral vertebrae and 4 unfused coccygeal vertebrae in their spines, which have 7 cervical vertebrae, 12 thoracic vertebrae, and 5 lumbar vertebrae. These sacral and coccygeal vertebrae gradually fuse together as a baby grows, giving rise to the 26 vertebrae that make up an adult spine. As an infant's body develops into adulthood, the number of vertebrae changes, which is a typical stage of development.

21. Correct Matching is: a-Q, b-R, c-P, d-S, e-T

a) Pivotal joint - Q (Neck)

One axis can be rotated around by a pivot joint. The head can rotate from side to side thanks to the neck's pivot joint.

b) Hinge joint - R (Elbow)

A hinge joint only permits movement in one direction, such as when a door is opened and closed. Arm flexion and extension are made possible by the elbow's function as a hinge joint.

c) Ball and socket joint - P (Shoulder)

Movement in all directions is possible with a ball and socket joint. As a ball and socket joint, the shoulder enables the arm to move in a variety of directions and lift, reach, and rotate.

d) Fixed joint - S (Cranium)

No movement is possible at a fixed joint. The bones of the skull are connected by a fixed joint called the cranium.

e) Ligament - T (Connects one bone to another)

A ligament is a tissue band that joins two bones together. Ligaments aid in stabilizing joints and limit excessive joint movement.

22. Correct Matching is: a-R, b-P, c-Q, d-T, e-S.

a) longest bone - R (Femur)

The longest bone in the human body, the femur, is the thighbone. It joins the hip to the knee and is situated in the upper leg. The femur is in charge of bearing the body's weight and enabling movement. The strongest bone in the body is the femur. It is composed of a dense, fracture-resistant network of bone tissue.

b) smallest bone - P (Stirrup)

The tiniest bone in the human body is called the stirrup. It is about 3 millimeters long, located in the middle ear. From the eardrum to the inner ear, sound vibrations are transmitted by the stirrup. The stapes is another name for the stirrup. It is one of the middle ear's three ossicles. The malleus and incus are the other two ossicles.

c) largest bird - Q (Ostrich)

The largest bird in the world is the ostrich. It can weigh up to 150 kilograms and reach heights of 2.7 meters. The only remaining species of the genus Struthio is the ostrich, which is native to Africa. The ostrich is a bird without wings, but it can run quickly. It is capable of travelling at up to 70 kph.

d) smallest bird - T (Hummingbird)

The world's tiniest bird is the hummingbird. It weighs less than 2 grams and can reach a length of up to 10 centimeters. Native to the Americas, hummingbirds can be found in a range of environments, including forests, mountains, and deserts. The hummingbird can hover in midair and flies extremely quickly. It is also an excellent pollinator and aids in the transfer of pollen from one flower to another.

e) Backbone - S (Vertebrae)

The row of bones that runs down the back is known as the backbone. It consists of 33 vertebrae stacked on top of one another. In order to connect the vertebrae, muscles and ligaments are used. The backbone supports the body's weight and permits movement. Additionally, it safeguards the spinal cord, a network of nerves that passes through the middle of the back.

LEVEL-II

1. Correct Option (B)

The organisms made up of a single cell are known as "unicellular," as correctly identified in option B. In unicellular organisms, there is only one cell, which is responsible for all vital life processes. Although these organisms' structures are frequently straightforward, they can be highly specialized and environment-adaptive. Protozoa, certain varieties of algae, bacteria, and some fungi are examples of unicellular organisms. As mentioned in option A, multicellular organisms, on the other hand, consist of numerous cells arranged into tissues, organs, and organ systems, allowing for increased complexity and specialization. In this situation, Option C, "Both A and B," is incorrect because it implies that the response could be either unicellular or multicellular, which it is not.

2. Correct Option (A)

Amoeba is an example of a unicellular organism, as indicated in option A. In unicellular organisms, a single cell performs all necessary life processes on its own. The protist amoeba is a well-known instance of a unicellular eukaryote. Amoeba has a straightforward physical structure, but it exhibits a variety of complex behaviors, including movement, feeding, and reproduction, all of which are carried out inside the boundaries of its single cell. Unicellular organisms like Amoeba are vital parts of the microbial world, playing a variety of roles in ecosystems and academic research. They can be found in a variety of environments, including freshwater, soil, and even inside the human body.

3. Correct Option (B)

The term used to describe organisms made up of more than one cell is "multicellular," as identified in option B. The characteristic of multicellular organisms is the presence of numerous cells arranged into tissues, organs, and organ systems. These cells collaborate to carry out specific tasks and add to the organism's overall structure and function. The evolution of multicellularity is a significant evolutionary change that has increased biological complexity and diversity and given rise to a diverse range of organisms, including fungi, animals, plants, and more. As mentioned in option A, unicellular organisms, on the other hand, are made up of just one cell that is capable of performing all vital life processes on its own. In this situation, Option C, "Both A and B," is incorrect because it implies that the response could be either unicellular or multicellular, which it is not.

4. Correct Option (C)

The correct answer is C. Plants and Animals. Animals and plants are both examples of multicellular organisms, which are made up of many cells that cooperate to form intricate structures. Plants can perform vital processes like photosynthesis and reproduction because their cells are arranged into tissues, organs (such as leaves, stems, and roots), and organ systems. On the other hand, animals display a variety of multicellular body plans as well as specialized tissues and organs for activities like digestion, respiration, and locomotion. Option C is the best option because it correctly identifies the two main groups of multicellular organisms in the biological world: plants and animals. These multicellular characteristics set plants and animals apart from unicellular organisms like Amoeba, as found in options A and B.

5. Correct Option (A)

The term used to describe the movement of animals from one place to another is "locomotion," as correctly identified in option A. Animals move in a variety of ways, or locomotion, including by walking, running, flying, swimming, or crawling. It is an essential part of an animal's life that allows them to forage and mate as well as explore their surroundings and avoid predators. Animals' ability to move around is a crucial adaptation that allows them to live and thrive in their respective ecosystems.

An organism's mobility is its capacity to move. It is a more inclusive term than locomotion and can refer to the movement of tissues, cells, and organs. Space between two points is measured as distance. It is not a substitute for movement.

6. Correct Option (D)

The statement is correct, as animals move for a variety of reasons, encompassing all the options provided. First off, animals frequently move in order to meet their basic needs for food and shelter. In order to locate suitable resources and habitats, they might need to explore various locations. Second, moving is an essential defense mechanism against enemies and predators. Animals have the ability to escape danger and improve their chances of surviving by hiding or running. Last but not least, movement is necessary for locating potential partners for reproduction. Many animals must find mates in order to breed, and this frequently entails moving to particular locations where potential partners can be found. In light of these various factors, the response "D. All the above" adequately encapsulates how important movement is to animals' survival strategies.

7. Correct Option (A)

The organ system that supports the body is the skeletal system, as mentioned in option A. The skeletal system is the body's intricate network of bones and cartilage, and it plays many important roles. Its main function is to support the body structurally, keeping the body in shape and safeguarding important organs like the heart, brain, and lungs. In addition, the skeletal system aids in movement by acting as points of attachment for muscles, allowing us to walk, run, and engage in a variety of activities. Additionally, it is essential for making blood cells and storing minerals, especially calcium and phosphorus. The skeletal system serves as the system that supports and underpins our physical form because it is crucial for the integrity and functionality of the human body as a whole.

8. Correct Option (B)

The animals with a backbone are known as vertebrates, as mentioned in option B. Vertebrates are a diverse group of creatures with internal skeletons that are fully developed and include a vertebral column or backbone. They differ structurally from invertebrates, which have no vertebral column. Vertebrates, which include numerous well-known animals like fish, mammals, birds, reptiles, and amphibians, occupy a variety of ecological niches on Earth. Their vertebral column is a distinguishing feature of this group because it provides stability, defense for the spinal cord, and the framework for the rest of the body.

9. Correct Option (A)

Animals without a backbone are known as invertebrates, as indicated in option A. Unlike vertebrates, which have a distinctive bony or cartilaginous vertebral column, invertebrates are a large and diverse group of animals. From simple organisms like sponges and jellyfish to more complex animals like insects, mollusks, and arachnids, this group contains a diverse range of species. Invertebrates play crucial roles in ecosystems all over the world by displaying a remarkable range of adaptations and occupying different ecological niches. Invertebrates are an intriguing and significant component of the animal kingdom because they rely on a variety of structural and support mechanisms as opposed to vertebrates, which have a clearly defined internal skeleton.

10. Correct Option (B)

The earthworm is indeed an example of an invertebrate, as stated in option B. Animals without a vertebral column or backbone are known as invertebrates. Earthworms are creatures with long, segmented bodies that are members of the phylum Annelida. They are a sizable portion of the enormous group of invertebrates that comprise most animal species on Earth. Vertebrates, on the other hand, are animals with a well-developed internal skeleton that includes a vertebral column or backbone, as was mentioned in option A. Because they lack this internal skeletal system, earthworms are classified as invertebrates. Earthworms should therefore be classified as invertebrates rather than vertebrates.

11. Correct Option (B)

An exoskeleton is a hard covering that protects the body of an invertebrate. It is constructed of the tolerant, flexible material chitin. Exoskeletons are present in most invertebrates. Insects, spiders, crustaceans, and mollusks are included in this. The exoskeleton supports the muscles, protects against predators, and helps to keep the body in shape. Invertebrates periodically shed their exoskeleton as they develop. This is known as moulting. The invertebrate can grow by moulting instead of shattering its exoskeleton. Invertebrates that lack an exoskeleton include worms and jellyfish. The soft bodies of these invertebrates are shielded by additional structures, such as a shell or mucus.

12. Correct Option (C)

Earthworms have tiny hair-like structures called "bristles," as mentioned in option C. The ventral surface of the earthworm's body is covered in these bristles, also referred to as setae. Chitin, the same substance used to create insect exoskeletons, is used to create the setae. They can be extended and retracted, and they are arranged in pairs along the earthworm's body. They are essential for the earthworm's movement and burrowing processes. The worm is held firmly in place by the bristles, which also help to grip and propel it through the tunnels it digs underground. Earthworms can move in a coordinated, wave-like motion by alternately contracting and relaxing the muscles associated with their bristles. This enables them to efficiently burrow through the soil, feed on organic matter, and carry out other crucial tasks.

13. Correct Option (B)

Snails have two pairs of tentacles, as mentioned in option B. On their head, snails have sensory organs called tentacles. Depending on the species, the eyes, which are carried by the upper pair of tentacles, can be straightforward or more intricate. The lower pair of tentacles primarily uses its senses of taste and touch to detect and explore its surroundings. These tentacles are essential for helping snails move about, find food, and detect potential dangers.

14. Correct Option (A)

The body of a cockroach is indeed divided into three parts: the head, thorax, and abdomen. The sensory organs, such as the eyes, antennae, and mouthparts, are housed in the head. The legs and wings (if any) are attached to the thorax, which is the middle section of the body. The digestive, reproductive, and respiratory systems are located in the abdominal region, which is the back of the body. Cockroaches and many other insects have anatomical structures that are distinguished by this three-part division. As a result, "Thorax," option A, is the right response.

The larynx, which is a component of the respiratory system, is absent from the cockroach's body. Tentacles are sensory organs, and cockroaches lack them on their bodies.

15. Correct Option (B)

Two sets of wings are attached to the breastbone of birds. The primary feathers are on the first pair of wings, and the secondary feathers are on the second pair of wings. The larger

and more significant of the two pairs, the primary feathers are used for flight. The secondary feathers, which are smaller, aid in balancing the bird while it is in flight. Feathers, which are joined to the wing's bones, make up a bird's wings. Lift and drag are produced by the arrangement of the feathers. Lift is the force that pushes the bird up into the air, and drag is the force that opposes the motion of the bird through the air.

The wings of a bird are an essential adaptation that allows them to fly. They are also used for balance and steering. Birds do not have two pairs of legs. They have one pair of legs, which are located on the underside of the body. The legs are used for walking, running, and perching.

16. Correct Option (A)

All vertebrates have a hard inner structure or framework supporting their body called a "skeleton," as indicated in option A. The skeleton is an integral part of the vertebrate body plan and performs a number of crucial tasks. It gives the body its shape and structural support, safeguarding important organs like the heart, lungs, and brain. The skeleton also provides a system of levers and serves as points for muscles to attach, which helps with movement. Additionally, it contributes to the development of red blood cells and the storage of minerals like calcium and phosphorus. While the precise makeup and organization of the skeleton may differ between different species of vertebrates, it is a characteristic that unites all of them.

The skeleton contains the flexible tissue known as cartilage. Although it is stronger than muscle, it is not as hard as bone. Cartilage is found in the joints, where it helps to cushion the bones and allow for movement. It is also found in the nose, ears, and larynx. Tendons are tough cords of tissue that connect muscles to bones. They help to transmit the force of muscle contraction to the bones, which allows for movement.

17. Correct Option (C)

Option C, "Fish," indicates that a fish has fins and a tail, which are specialized adaptations that allow it to swim effectively inside water. Fish can change direction, maintain balance, and modify their depth thanks to the stability and control that fins give them in the water. The primary fish propelling mechanism is the tail, also known as the caudal fin. It generates the thrust required to move forward in an aquatic environment through its back-and-forth motion. These characteristics make fish well-suited for life in aquatic habitats, whether it be in oceans, rivers, lakes, or ponds. They also help them navigate through water with ease and precision.

18. Correct Option (C)

A fish's skeleton is covered in muscles. Fish have an internal framework of bones or cartilage called an endoskeleton, which serves as structural support for the fish's body. A fish's ability to move and move around is greatly influenced by its muscles, which are attached to its endoskeleton. Fish can swim, change directions, and perform a variety of other movements by flexing their skeletal components by contracting and relaxing their muscles. Therefore, a fish's ability to move and control itself depends on the muscles that cover its endoskeleton.

19. Correct Option (D)

Snakes have rectangular scales that line the underside of their body, making option D, "Snake," the correct answer. The snake's locomotion is greatly aided by these scales, also known as ventral scales or belly scales, which are distinct from the scales on the snake's back. These belly scales are arranged in a smooth, overlapping pattern that gives snakes good traction and grip on a variety of surfaces. They are well adapted to their serpentine way of life thanks to this particular scale arrangement, which enables them to slither across the ground, climb trees, and navigate their surroundings.

20. Correct Option (B)

Birds have a spindle-shaped (streamlined) body which reduces air resistance. Air resistance is the force that opposes the motion of an object through the air. A body is said to be streamlined if its surface is smooth and curved, as this lowers air resistance. A bird's spindle-shaped body reduces air resistance in two ways. First, the body's curved, smooth surface aids in reducing turbulence in the air flow surrounding it. The air moving erratically in turbulence causes drag. Second, the spindle-shaped body helps to minimize drag by directing air flow over the body. Having less air resistance allows birds to fly more effectively. This is due to the fact that flying a bird uses less energy. Birds with spindle-shaped bodies can therefore fly for longer stretches of time and cover greater distances.

21. Correct Option (A)

The upward movement during the flapping of wings is called the "upstroke," as mentioned in option A. A continuous cycle of wing movements is used by birds to produce lift and thrust during flight. As it gets ready for the upcoming downstroke, the bird's wing moves upward during the upstroke, usually against the resistance of the air. In order to maintain aerodynamic stability and control during flight, the upstroke is essential. The wing then moves downward during the downstroke, producing the main force required for lift and propulsion. Together, these back-and-forth up-and-down movements produce the necessary airflow patterns for birds to fly effectively and navigate through the air.

22. Correct Option (B)

Birds' strong chest muscles play a key role in the up-and-down movement of their wings during flight. In order to produce the required force for flapping their wings, birds have highly developed pectoral muscles. The lift and thrust needed by birds to take flight and maintain altitude are produced by the up-and-down motion of their wings. One of the key adaptations that allows birds to achieve powered flight and perform a variety of aerial manoeuvres is the development of these specialised chest muscles. Many animals, in contrast to birds, lack these specialized wing muscles, which prevents them from being able to fly in the same way that birds can.

23. Correct Option (B)

When a bird lands, perches on a branch, or walks on the ground, its tail feathers are crucial in helping it stay balanced. The tail, which is made up of all these feathers, stabilizes the bird's body while it engages in various activities. The tail feathers of a bird aid in directing and stabilizing its flight. But even when perching, landing, or walking on the ground, the tail feathers balance the animal by changing their position and angle as necessary. The ability to maintain balance is necessary for the bird to maintain its posture, avoid falls, and perform tasks with accuracy, making the tail feathers an important part of a bird's anatomy.

24. Correct Option (C)

Penguins, particularly those found in Antarctica, have wings that have been modified into flipper-like structures that are ideal for swimming. Although penguins are flightless birds and therefore cannot fly in the conventional sense, their wings have developed to function as strong underwater flippers. They can swim through the icy waters of the Antarctic region with great agility and efficiency thanks to their modified wings, pursuing and catching prey like fish and krill. Because they spend a large portion of their lives in the ocean and depend on their specialized wings to survive in their aquatic environment, penguins must be able to swim in order to survive.

25. Correct Option (D)

The answer is D. 206 and 600. An adult human body has 206 bones and over 600 muscles. The bones form the skeleton, which provides support and structure for the body. The muscles are attached to the bones and allow for movement.

Everybody's body contains a slightly different number of bones. Due to birth defects or injuries, some people may have more or fewer bones. However, an adult human body has 206 bones on average.

The human body's muscles are also variable. Because of their genetic makeup or level of physical activity, some people may have more or fewer muscles. However, an adult human body has more than 600 muscles on average.

Together, the bones and muscles support and enable movement in the body. The muscles attach to the bones as a framework, and they contract and relax to move the bones. The bones also protect the organs and tissues of the body.

26. Correct Option (B)

The right response is option B, "22," because the human skull is made up of 22 bones. The cranium, which houses and shields the brain, and the facial bones, which make up the structure of the face, are two of these 22 bones. The brain and other sensory organs, like the eyes and ears, are protected by the skull, which also serves as the framework for the face's features. The skull is a crucial component of the skeletal system.

The portion of the skull that houses the brain is known as the cranium. There are 8 bones in it:

Occipital bone, Sphenoid bone, Ethmoid bone, Frontal bone, Parietal bone (2), Temporal bone (2), and Parietal bone (2)

The bones that make up the face are the facial bones. They are made up of 14 bones:

Maxilla (2), Mandible, Zygomatic bone (2), Nasal bone (2), Lacrimal bone (2), Palatine bone (2), Inferior nasal conchae (2), Vomer, Inferior turbinate

27. Correct Option (C)

Option C, "33," is the right response because the vertebral column is made up of 33 ring-shaped bones called vertebrae. The spinal column, which runs down the back and gives the spinal cord vital support and security, is made up of these vertebrae stacked on top of one another. The cervical (7 vertebrae), thoracic (12 vertebrae), lumbar (5 vertebrae), sacral (5 fused vertebrae), and coccygeal regions of the vertebral column are each comprised of five vertebrae (4 fused vertebrae). Together, these vertebrae function as an essential component of the human skeletal system because they safeguard the spinal cord while also enabling flexibility and movement.

28. Correct Option (D)

The ribcage is a cage-like structure of bones that protects the heart and lungs. The 24 ribs that make up this structure are joined to the sternum (breastbone) in the front and the spine in the back. Although the ribcage is strong enough to shield the heart and lungs from harm, it is also flexible enough to allow for breathing.

Blood is pumped throughout the body by the muscular heart. It is situated directly behind the sternum in the middle of the chest. We can breathe thanks to the lungs, which are two spongy organs. They are situated on either side of the heart, and the mediastinum and heart are what divide them (the space between the lungs).

By absorbing shock and avoiding crushing them, the ribcage shields the heart and lungs from harm. It also helps to keep the lungs expanded, which is necessary for breathing.

29. Correct Option (C)

The limbs' bones are supported by girdles. Girdles are bodily parts of people that attach the limbs to the axial skeleton (the central part of the skeleton that includes the skull, spine, and ribcage). The pectoral girdle, which joins the upper limbs (arms) to the axial skeleton, and the pelvic girdle, which joins the lower limbs (legs) to the axial skeleton, are the two main girdles in the human body. These girdles are essential for supporting and stabilizing the limb bones, which enables a variety of movements and functions like walking, running, and lifting. Therefore, girdles are vital for the proper functioning and mobility of the limb bones, making option A, "Limbs," the correct answer.

30. Correct Option (C)

The longest bone in the human body, which is located in the thigh, is called the "femur." The main bone of the upper leg is the femur, also known as the thigh bone, which runs from the hip joint to the knee joint. Its length and strength are crucial for giving the lower body support, stability, and mobility. Its length and strength make it essential for maintaining stability and balance while moving through various physical activities like walking and running.

The flexible tissue that cushions joints is cartilage. Tendons are strong tissue cords that attach muscles to bones. Tendons and cartilage are not bones.

8. Adaptation of Animals to Climate
Solutions

LEVEL-I

1. Correct option: (A) Very hot and wet

The climate found in Kerala is categorized as "very hot and wet". Accordingly, Kerala has high temperatures and humidity all year round. The area has high rainfall throughout the monsoon season, which adds to its profusion of wildlife and beautiful foliage. The distinctive natural beauty and picturesque landscapes of Kerala are mostly due to its warm and humid climate, which also makes it a great location for a variety of plant and animal species to thrive.

2. Correct option: (A) In the afternoon

When the sun rises, the temperature reaches its peak for the day. The Earth's surface starts to warm when the sun rises, which raises temperatures throughout the morning and early afternoon.

Although the maximum temperature is normally measured around midday when the sun is at its highest point in the sky, this heating impact peaks in the late afternoon. After then, when the sun sets and the Earth radiates heat away, the temperatures begin to steadily fall. Thus, the hottest time of the day often occurs in the morning, around noon, before it begins to drop off in the evening.

Early in the day, when the sun is low in the sky and its rays are less direct, the temperature is at its lowest of the day. The air above the Earth's surface has cooled down throughout the course of the previous night.

3. Correct option: (B) Sand cools down faster

Nights in deserts are much cooler primarily because of the low heat-retaining capacity of sand. During the day, the desert sand absorbs a significant amount of heat from the sun and becomes very hot. However, as the sun sets and the desert loses its direct exposure to sunlight, the sand cools down rapidly due to its low heat retention properties.

This leads to a substantial drop in temperature, making desert nights considerably cooler compared to the scorching daytime temperatures. Option (b) correctly identifies this phenomenon as the reason for cooler nights in deserts.

The other choices are incorrect. In deserts, the humidity does not rise significantly at night. Although wind does blow in deserts, it does not blow hard enough to considerably chill the air. Deserts have no nighttime precipitation.

4. Correct option: (A) Climate

Climate is the annual average of a location's long-term average temperature and precipitation. A region's climate is made up of the regular weather factors that prevail there for a prolonged period of time, often 30 years or more, including temperature, precipitation, humidity, and other atmospheric components.

It gives us insight into the typical weather patterns and trends in a region, enabling us to distinguish between various climatic zones like tropical, temperate, dry, etc. Climate is the proper identification of this phrase in Option (a).

The other choices are incorrect. Climate is the long-term average of a region's weather conditions. The state of the atmosphere at a specific location and time is called the weather. The technique of anticipating future weather conditions is called weather forecasting.

5. Correct option: (D) All of these

Temperature, humidity, and rainfall are a few examples of weather components. The term "weather" describes the transient atmospheric conditions and changes that take place

from day to day or even hour to hour. These factors are very important in influencing the local weather at any given time. Other weather factors may include cloud cover, visibility, wind speed, air pressure, cloud cover, and atmospheric conditions like fog or frost. Option (d) is correct as it includes all the mentioned elements of weather.

6. Correct option: (D) All of these

Forecasting the weather has many uses and benefits in daily life. By giving people early warnings and enabling them to take the required steps to preserve lives and property, it assists in preparing for adverse weather conditions, such as storms, hurricanes, or excessive rains.

It helps with daily activity planning since it allows people to plan outside events, travel, and other activities in accordance with the weather prediction. Weather forecasting is also essential for organizing agricultural activities since it allows farmers to arrange irrigation, planting, and harvesting operations depending on the expected weather. The right response is option (d), which contains all of the advantages of weather forecasting that have been highlighted.

7. Correct option: (D) All the above

The humidity of a place refers to the amount of water vapor present in the air. It is an indicator of the amount of moisture in the air and has a big impact on how comfortable or uncomfortable people feel in different weather situations. Low humidity can result in dryness, which can have an adverse effect on the health of the skin, the respiratory system, and some materials. High humidity can make the air seem heavy and stuffy.

Precipitation, or falling water from the sky, includes both rainfall and snowfall. Humidity and precipitation are not the same. Precipitation is the amount of water that falls from the sky, whereas humidity is the amount of water vapor in the air.

8. Correct option: (C) Polar region

Penguins have evolved to exist in the polar regions, especially in Antarctica and its surroundings. They are equipped to endure in severely cold and icy conditions. Penguins have various adaptations that allow them to survive in these extreme temperatures, including a coat of insulating blubber, thick feathers that trap air for further insulation, and unique glands to remove extra salt from their bodies. Their distinctive body composition and streamlined form also help them swim and dive more effectively in frigid waters. The polar area is where penguins are evolved to reside, hence option (c) is the right response.

Equatorial areas are hot and humid, whereas rainforests are warm and moist. Deserts are hot and dry. Because they are not suited to the heat or the humidity, penguins would not be able to live in these environments.

9. Correct option: (B) Rajasthan

The state of Rajasthan, which lies in the northwest of India, is renowned for its hot and dry weather. It frequently encounters summertime temperatures far beyond 40^0 Celsius (104 degrees Fahrenheit), which are considered to be severe. The area is known for its extremely low rainfall and dry and semi-arid climate. Rajasthan is among the hottest and driest places in India due to its climate. The Thar Desert, the largest desert in India, is located in Rajasthan.

The other choices are incorrect. The climates of Kashmir are chilly and mountainous, Kerala is hot and humid, and the North East has a humid, subtropical environment.

10. Correct option: (B) They cannot swim

Penguins are well-suited to live in the water and have excellent swimming abilities. Being excellent swimmers, they move through the water with tremendous agility using their wings, which have changed into flippers. Penguins spend a large amount of their lives in

the water where they forage for fish and other marine life. They are widely recognized for their swimming prowess.
The other statements are all correct. Penguins have webbed feet, streamlined bodies, and do snuggle together. Penguins keep warm by huddling together, and their webbed feet and sleek bodies enable them to swim and dive.

11. Correct option: (C) Desert regions

The adaptive traits listed in the query, such as being nocturnal, having big, cushioned feet, sweating less, and having water-saving systems, are traits frequently observed in animals living in desert environments. Animals living in deserts may survive in their environment despite the intense heat and lack of water because to these characteristics.
Numerous desert animals are nocturnal, meaning they are most active at night when it is cooler and less water evaporates. Desert animals are able to avoid sinking into the sand because to their wide, cushioned feet, which help them disperse their weight over a broader area. Because sweating causes water loss and desert animals must preserve water, they do not sweat as much. Some desert animals have unique water-saving strategies to aid them. For instance, camels can create water when necessary thanks to the fat that is stored in their hump.

12. Correct option: (A) Climate

The average weather pattern taken over a long period, such as 25 years or more, is known as "climate." The term "climate" refers to the normal atmospheric conditions, such as temperature, precipitation, humidity, wind patterns, and other meteorological factors, that are long-term fixtures in a certain area. It helps us comprehend the typical weather patterns encountered in many parts of the world and gives us a larger perspective on the long-term weather trends in a certain place.
Weather refers to the atmosphere's present condition at a specific location and time. The process through which a species or creature adapts to its environment over time is change. Humidity refers to how much water vapor is present in the atmosphere.

13. Correct option: (A) Polar region

The coldest region on Earth is the polar region, specifically the areas near the North Pole and the South Pole. Due to their proximity to the Earth's poles and the prolonged darkness of the winter, these areas suffer very frigid temperatures all year long. The main polar regions recognized for their arctic climates, ice, and snow are Antarctica and the Arctic, both close to the poles.
On the other hand, tropical locations are close to the equator and receive the most direct sunshine, which causes the temperature to rise as well. The temperate zones, which lie in between the tropical and polar regions, are more temperate than any of them. Near the oceans are the coastal zones. Coastal places often have milder temperatures than inland areas because the seas serve to regulate local temperatures there.

14. Correct option: (A) Fatty tissue

The hump of a camel is a reservoir of fatty tissue, not water or milk. The capacity of camels to store fat in their humps, which they may use as an energy reserve during periods of food scarcity, is widely documented. Contrary to common opinion, camels are extremely adept at conserving water and may go for extended periods without drinking. Instead of acting as a water storage organ, the hump. During its travels through dry and desert areas, the camel receives vital nutrition and energy from the fat tissue in the hump. The camel's bodily fluids, not its hump, are where it stores water. The kidneys of the camel are excellent at retaining water. The camel's mammary glands, not its hump, are where milk is produced.

15. Correct option: (A) Pressure

The appropriate response is (a) Pressure. Usually, a daily weather report does not include pressure. In order to convey information about the present or predicted weather conditions, the daily weather report often concentrates on factors like Temperature (b), Rainfall (c), and Humidity (d). Understanding the current atmospheric conditions and making appropriate plans for your everyday activities depends on these factors.

Although crucial for analyzing weather patterns and formulating forecasts, pressure is not often covered in a brief and uncomplicated daily weather report intended for the general audience. Instead, to evaluate and forecast more complicated weather systems, meteorologists frequently study and analyses pressure values and trends.

16. Correct option: (C) Meteorological department

The department that prepares the weather report is the meteorological department. It is in charge of tracking, evaluating, and predicting weather patterns. To provide precise and fast weather reports and predictions, the meteorological department gathers data from numerous weather stations, weather satellites, and other sources. These reports are essential for many industries that rely on weather data, including public safety, aviation, agriculture, and many others.

A nation's telecommunications infrastructure, including phone lines, cell towers, and internet cables, is the responsibility of the telecom department. A nation's electrical system, which distributes electricity to residences and businesses, is under the control of the electrical department. Buildings, bridges, and other public works projects are within the purview of the civil department.

17. Correct option: (D) Moisture

The amount of humidity in the atmosphere is a measure of the moisture content in the air. In order to evaluate weather conditions and determine how comfortable it is for people to be outside, it is crucial to know the air's humidity level. Our sense of temperature can be affected by the air feeling damp and high humidity levels. On the other side, low humidity can result in dry air, which can have an adverse effect on the skin, respiratory health, and some materials.

To determine how hot or cold something is, use the term "heat." A substance's temperature is a measure for the amount of heat it contains. Materials comprised of atoms or molecules are known as chemicals.

18. Correct option: (B) Rain gauge

The instrument used to measure rainfall is called a rain gauge. It is a simple meteorological instrument made to gather and measure the total amount of precipitation—mostly rainfall—over a predetermined time frame. Although there are many other types of rain gauges, the most popular one is a cylindrical container with markings showing the quantity of rainfall in inches or millimeters. Rain comes into the gauge, gathers in the container, and the height of the water gathered reflects the quantity of precipitation that fell during the specified time period.

An instrument used to gauge atmospheric pressure is a barometer. An instrument used to measure temperature is called a thermometer. A maximum-minimum thermometer is a device that measures the temperatures at their highest and lowest points throughout a specific time period.

19. Correct option: (C) Maximum – minimum thermometer

The maximum and minimum temperatures of a day are recorded using a maximum-minimum thermometer. A 24-hour day is typically the time span for which this sort of thermometer is intended to monitor and record the peak and lowest temperatures. Two distinct scales—one for monitoring the highest temperature and the other for the

minimum temperature—and a U-shaped glass tube make up the device. The liquid expands as the temperature rises and travels up the tube, revealing the highest temperature achieved. The liquid, on the other hand, contracts as the temperature decreases and flows down the tube, indicating the minimum temperature.
Both a Celsius and a Fahrenheit thermometer are used to monitor temperature, however neither one records the peak and lowest temperatures for a certain time period. Although a hydrogen thermometer may measure extremely low temperatures, it is uncommon to use one to note the high and low temperatures for a given day.

20. Correct option: (B) Sun

All the changes taking place in the atmosphere are primarily caused by the (b) Sun. The fundamental energy source for the Earth's weather and climate systems is the Sun. The uneven heating of the Earth's surface caused by solar radiation leads to the creation of air masses, wind patterns, and weather events. The water cycle, which produces clouds and precipitation through the evaporation and condensation of water vapor, is likewise fueled by the energy of the Sun.
The atmosphere's changes are not significantly influenced by the Moon, Earth, or stars. The tides are impacted by the Moon's gravity, although the atmosphere is not significantly affected. There are minor changes in the atmosphere brought on by the Earth's rotation, although these changes are minimal. The atmosphere is not much impacted by the stars at all.

21. Correct option: (A) Polar regions

The places where the sun does not rise for six months and does not set for the other six months are the (a) polar regions. These areas are close to the poles of the Earth, notably the North and South Poles. The tilt of the Earth's axis causes continuous darkness for roughly six months during the polar winter and continuous sunshine for about six months during the polar summer. The Earth's axis is tilted in relation to its orbit around the Sun, which affects the length of the day at various latitudes throughout the year.
This phenomenon is not present in the equatorial, northern, or southern areas. Even though the amount of sunshine varies throughout the year, the sun rises and sets in these areas every day.

22. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.
Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

23. Correct option: (D) Weather

he short-term variation that occurs in the atmosphere is called "weather." The daily variations in temperature, humidity, precipitation, wind speed, and cloud cover that can occur over short periods of time are referred to as weather. It is the state of the atmosphere at a specific location and time, offering details about the current circumstances.
Contrarily, climate refers to the typical weather patterns for a certain place over an extended period of time. Usually, the length of a climate measurement is 30 years. The quantity of water vapor in the air is known as humidity. What determines how hot or cold anything is its temperature.

24. Correct option: (B) altitude or height above sea-level

The climate at a place depends on various factors, but one of the most significant factors is altitude or height above sea-level. The atmospheric pressure drops as altitude rises, resulting in thinner air. Weather patterns and temperature are directly impacted by this alteration in atmospheric circumstances. In general, temperatures are cooler at greater altitudes, which explains why mountains and high plateaus typically have colder climates than low-lying locations.

While elements such as air pressure, wind speed, and temperature affect climate in crucial ways, the height or elevation above sea level frequently has a decisive impact on the general climatic features of a given region.

25. Correct option: (D) Polar bear

The animal that possesses blubber under its skin is the (d) Polar bear. Blubber is a thick layer of fat found under the skin of certain marine mammals, such as whales, seals, and polar bears. It serves as an insulating layer, helping these animals to stay warm in cold environments, particularly in the frigid waters of the polar regions. Polar bears, in particular, rely on their blubber to maintain body heat and survive in the Arctic's harsh and icy conditions.

There is no blubber under the skin of red-eyed frogs, elephants, or birds. In tropical jungles, there exist tiny amphibians called red-eyed frogs. In savannas and woodlands, elephants are huge creatures. Feathered creatures with wings are birds. Due to their climatic adaptations, none of these creatures require blubber to keep warm.

LEVEL-II

1. Correct option: (C) Toucan

A big, long beak is one of the distinguishing characteristics of toucans, which is known about. In addition to its amazing size, its beak has rich coloring that is frequently decorated with vivid colors. In addition to other functions, a toucan's beak enables it to reach fruits and seeds that are higher up on branches than it is physically able to reach. It also participates in communication, territorial displays, and assists in mate attraction.

Pigeons, penguins, and kiwis don't have very lengthy beaks. In order to dig for food, kiwis have long, narrow beaks. The small, stubby beaks of penguins are employed for capturing fish. Pigeons utilize their short, pointed beaks to devour seeds and other small foods.

2. Correct option: (D) Catch its prey

A sharp beak is a specialized weapon used by eagles to effectively grab and eat their prey. Eagles are raptors with exceptional hunting skills, and their beaks are built to shred and tear apart their prey's flesh. They can grip and hold onto their catch tightly thanks to their sharp, hooked beak.

Additionally, it helps the eagle swiftly break off bite-sized chunks of meat, making it simpler for it to devour. While eagles occasionally migrate, they are not known for hiding from or fleeing from predators, and their beaks are mostly specialized for their carnivorous diet and hunting ability.

They are apex predators, meaning they are at the top of the food chain and lack natural predators, therefore option A is to flee from them. Because they are huge birds, they are not particularly adept at hiding from predators, thus choose option B. Eagles migrate, option C, although their beak does not aid them in this endeavor.

3. Correct option: (D) All of these

Numerous elements, such as average temperature, soil type, and yearly precipitation, have an impact on the variety of species in varied ecosystems. Each of these elements has a significant impact on the habitat's environmental characteristics, which in turn impacts

the kinds of species that may survive there. Different species have distinct adaptations that enable them to endure and procreate in certain environmental circumstances. For instance, certain species could do better in hot, dry habitats with particular soil types, whereas others might do better in cold, damp conditions with various soil traits. The sum of these elements leads to the total biodiversity seen in different environments across the world.

4. Correct option: (B) Orchids

The term "epiphyte" refers to plants that grow on top of other plants (often trees) without taking nutrients from those plants. The most popular epiphytes are orchids. With the aid of specialized adaptations, they may adhere to tree trunks or branches and draw water and nutrients from the surrounding trash, the rain, and the air.

The popularity of orchids in horticulture and decorative plant collections can be attributed to their well-known for having an array of exquisite blossoms. The well-known tree species fir and pine do not develop as epiphytes on other plants, despite the fact that they are not epiphytes.

Both fir and pine are tree species, not epiphytes. They are rooted to the ground and develop roots as they grow in the soil.

5. Correct option: (B) Hibernation

Hibernation is the term for when animals sleep or go into a state of reduced activity in the winter to preserve energy and endure severe weather. Many animals, especially tiny mammals like bears, groundhogs, and bats, hibernate as a seasonal adaptation to withstand lower temperatures and scarcer food sources. The animal may use its stored energy stores to survive until the warmer months return since during hibernation its metabolic rate and body temperature drop dramatically to fit the environment. Torpor enables animals to endure the harsh winter months when food is in short supply.

Different methods that animals adapt to their surroundings include camouflage, movement, and mimicry. With the use of camouflage, an animal can conceal itself from predators by blending in with its environment. An animal can move about its surroundings through locomotion. Animals can mimic other organisms to look like them, frequently for trickery.

6. Correct option: (D) All of these

In addition to India, Malaysia is located in Southeast Asia, Brazil is in South America, Kenya is in East Africa, the Republic of Congo is in Central Africa, Indonesia is in Southeast Asia, Nigeria is in West Africa, and Uganda is in East Africa. Warm temperatures, high humidity levels, and year-round rainfall in these areas often provide the perfect environment for the luxuriant and diversified ecosystems seen in tropical rainforests. Numerous plant and animal species found nowhere else on Earth may be found in these jungles, which are renowned for their extraordinary biodiversity

7. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.

Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

8. Correct option: (B) Camouflage

A turtle may camouflage itself by blending in with its surroundings using the colors and patterns on its shell. The ability to blend in with their environment allows animals of many different kinds to use camouflage, which makes it more difficult for predators to identify them and for prey to detect their presence.
In the case of turtles, the patterns and hues of their shells frequently match the forest floor or other aspects of their natural habitat, giving them some kind of defense against possible dangers. By lowering their probability of being seen by predators or prey, this adaptation boosts their odds of surviving.
Some creatures go into hibernation during the winter months, which is a condition of inactivity and low metabolic rate. Animals travel from one location to another during migration, frequently in pursuit of food or a better environment. Behavior is how an animal acts or responds to its surroundings.

9. Correct option: (C) Average in weather pattern in a few years

Climate is the long-term average of the weather in a certain area over a lengthy period of time, usually spanning decades or even centuries. It considers variables including temperature, humidity, precipitation, wind patterns, and other long-term atmospheric conditions. For a variety of uses, including agriculture, urban planning, and environmental research, climate provides important information that helps us understand the normal or predicted weather patterns in a particular place.
On the other hand, weather is the momentary condition of the atmosphere. From day to day, or even from hour to hour, it might alter. Many of the same elements that affect climate also affect weather, although it is also impacted by more regional causes like thunderstorms and hurricanes.

10. Correct option: (A) A white fur, fat below skin, keen sense of smell

Polar bears can survive in extremely cold settings because to a number of adaptations, including:
Their fur is really translucent, giving it the appearance of being white, which helps them conceal themselves when hunting and blend in with their snowy surroundings.
Polar bears have a thick layer of blubber under their skin that serves as insulation and aids in the preservation of body heat in the chilly Arctic conditions.
Polar bears have a keen sense of smell that enables them to locate seals and other prey at a distance. This skill, which enables them to find food even when it is concealed beneath the ice, is essential to their survival in their frigid habitat.
The alternative choices are incorrect. Large eyes, thin skin, and a long tail would all be problems for bears in the cold. Large eyes would make it difficult for bears to see in the snow. The bears' huge, white paws would aid in swimming but not in keeping them warm.

11. Correct option: (B) Swimming in water

Frogs and ducks have webbed feet as a form of adaptation to help them swim effectively in water. Their feet have larger surface area thanks to the webbing between their toes, enabling them to push against the water with more power. This helps them swim and move through watery surroundings. For ducks and aquatic frogs who spend a lot of time in the water, this adaption is extremely advantageous.
The other options are not correct. Frogs and ducks do not need their webbed feet to hold branches tightly, fly in the air, or grab and eat insects.

12. Correct option: (C) Adaptation

An organism's ability to grow and live in a given climate or habitat is referred to as adaptation. These characteristics are the product of natural selection and evolutionary processes, in which qualities that offer an organism a competitive edge in a particular environment is favored and handed down through generations. In order to properly obtain

food, shelter, and reproduce within their ecological niche, organisms undergo adaptations to deal with elements including temperature, precipitation, humidity, and other environmental circumstances. Therefore, an organism's fitness and success in a given environment are greatly influenced by its adaptations.
The alternative choices are inaccurate. Climate variables including temperature, humidity, and wind are referred to as the short-term weather. Climate is the collection of long-term, typical weather patterns. None of these words discuss the specific traits that an organism possesses that make it suitable to a certain environment.

13. Correct option: (A) 15 º – 40 º C

The temperature in the tropics ranges from 59 º to 104 º Fahrenheit, or 15 º to 40 ºC. Near the equator, tropical zones receive warm to hot temperatures all year long. Due to these areas' close closeness to the equator, where they receive constant, bright sunshine, the temperature there stays relatively high. Depending on the location, altitude, and other local conditions, the precise temperature within this range might change, but tropical locations often have warm climates with high humidity levels that sustain a wide variety of ecosystems and lush flora.

14. Correct option: (B) Arboreal fi burrow

Burrows and arboreal habitats are two separate categories of animal habitats. Burrowing animals dwell underground in tunnels or shelters, whereas arboreal creatures live in trees. Some creatures, like squirrels, may dwell in both burrows and trees, therefore these two habitat types are not mutually exclusive. The pair Arboreal fi Burrow is incorrect since the two phrases are not equivalent.
The other choices are the right ones. Animals classified as terrestrial or aquatic or amphibian live on land, in water, or both.

15. Correct option: (C) Long grasping tail

All the other choices (A, B, and D) are traits that are typically associated with creatures that are evolved to surviving in cold conditions, such as the Arctic or polar areas, which is why C) is the odd one out.
A) Thick layer of fat under the skin: Polar bears and seals use this adaption, known as blubber, to help them insulate themselves.
B) White fur: As a kind of camouflage, several animals have evolved white fur to assist them blend in with their environment.
D) Animals with broad, big feet and keen claws, such as polar bears, frequently exhibit this adaptation, which makes it easier for them to move around and grasp frozen surfaces and to successfully acquire prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

16. Correct option: (B) They cannot swim

Penguins have exceptional swimming skills, making them well-suited to life in the water. They are adept swimmers and use their wings, which have become flippers, to navigate through the water with great dexterity. Penguins spend a significant portion of their lives foraging for fish and other aquatic creatures in the ocean. They are also known for having excellent swimming skills.
All of the other claims are true. Penguins do cuddle up together and have webbed feet and streamlined bodies. Penguins can swim and dive thanks to their webbed feet and streamlined bodies, and they keep warm by huddling together.

17. Correct option: (C) To blend in and hide

The ability to change color allows chameleons to blend in with their environment. This enables them to ambush prey and hide from predators. They can adjust their skin color to match the colors of nearby objects, such as leaves or branches, which helps them remain unnoticed. Chromatophores are peculiar skin cells that are present in chameleons. These cells carry pigment in a variety of colors. These cells are really growing or shrinking as a chameleon changes color. This alters how much pigment is exposed, which in turn alters the skin color of the chameleon.
The other options are not correct. Chameleons do not change color to use less energy, to change their temperature, or to help them hibernate.

18. Correct option: (A) Highest day temperature and lowest night temperature of the day

The highest recorded temperature during the day, often in the afternoon, is referenced by the maximum temperature that is shown in the weather broadcast. The highest temperature that day was represented by this.
The minimum temperature listed in the weather broadcast is the lowest reading made overnight or early in the morning, usually just before daybreak. The lowest temperature that day was represented by this.
The alternate choices are incorrect. Usually, the weather bulletin does not include the month's highest daytime or nighttime temperatures. Additionally, it is not customary to include the temperature recorded at noon and at midnight because these are not necessarily the highest and lowest temperatures of the day. The average highest and lowest temperatures of the day and night are also not often shown since they are less useful than the actual highest and lowest temperatures.

19. Correct option: (C) Red-eyed frog and beard ape

Red-eyed frogs and beard apes are both found in tropical rainforests, which are hot and humid environments. The rainforests provide these animals with the moisture and shade they need to survive.
All around the world, tropical rainforests are home to red-eyed frogs. They can see in the dark thanks to their vivid red iris. Additionally, sticky pads on the foot of red-eyed frogs enable them to climb trees.
The Western Ghats of India are the home of the bearded ape. They go by the name "lion-tailed macaques." A long, thick mane of hair in bearded apes serves as sun protection. Additionally, they have a long tail that aids in balance while climbing trees.
Elephants in Africa and polar bears can only be found in cold regions; hence they cannot live in hot and humid settings. The tundra and desert, which are both cold and dry habitats, are where you may find camels and reindeer.

20. Correct option: (C) Average weather pattern of many years

Climate is the long-term, typically spanning decades or even centuries, average of the weather in a particular place. Temperature, humidity, precipitation, wind patterns, and other long-term climatic conditions are all factors that are taken into account. Climate offers crucial information that helps us comprehend the typical or anticipated weather patterns in a certain location for a range of tasks, including agriculture, urban planning, and environmental research.
Contrarily, weather refers to the current state of the atmosphere. It might change from day to day, or even from hour to hour. The same factors that influence climate also have an impact on weather, albeit it is also influenced by more localized factors like thunderstorms and hurricanes.

21. Correct option: (D) All of these

Tropical rainforests are characterized by hot and humid climates, enormous numbers and varieties of animals, and competition for food and shelter.
The environment in tropical rainforests is often hot and humid, with average temperatures of 15° – 40° C. Additionally, there is a lot of humidity, with typical percentages ranging from 70 to 90%
Huge variety and quantity of creatures: The tropical rainforests are home to a huge diversity and number of animals. Tropical rainforests are home to about half of all species on Earth, or an estimated 10 million different kinds of plants and animals.
Competition for food and shelter is a result of the high levels of biodiversity seen in tropical rainforests. These creatures may evolve specialized adaptations as a result of this intensive competition.

22. Correct option: (A) They cannot swim

Penguins can swim quite effectively, which makes them perfect for a life in the sea. They are skilled swimmers and can navigate through the water with incredible agility using their wings, which have evolved into flippers. Penguins forage on fish and other aquatic animals in the ocean for a sizable portion of their life. They are also renowned for their prodigious swimming abilities.
The other assertions are all accurate. Penguins have webbed feet, streamlined bodies, and they do snuggle up together. Penguins' webbed feet and streamlined bodies enable them to swim and dive, and they keep warm by huddling together.

23. Correct option: (C) Minimum and maximum thermometer

A unique kind of thermometer known as a minimum-maximum thermometer is used to track the daily minimum and maximum temperatures. There are two bulbs in it, one loaded with mercury and the other with alcohol. The minimum temperature is recorded using an alcohol bulb, while the highest temperature is recorded using a mercury bulb.
The temperature of a person or a material is measured using clinical thermometers or laboratory thermometers, respectively. They are not intended to keep track of the daily minimum and maximum temperatures.

24. Correct option: (D) Tropical rainforest

A tiger is a carnivorous animal that fits the definition; it is distinguished by its striking stripes and extraordinary speed and agility when pursuing prey. In Asia, grasslands and tropical rainforests are the main habitats of tigers. They are strong predators that have evolved to live in these luxuriant, thick forest habitats. These animals have stripes on their body that help them to camouflage in the dense vegetation of the rainforest. They also have sharp claws and teeth that help them to catch their prey.
The Arctic, which is significantly dissimilar to the tropical jungle in terms of habitat, is where polar bears may be found. Tropical rainforests are often very hot and humid, while deserts are typically very hot and dry. Ocean-dwelling animals have gills or a thick covering of blubber as adaptations to help them survive in the saline water.

25. Correct option: (C) Long grasping tail

The qualities of all the other options (A, B, and D) are often associated with animals that have evolved to survive in frigid climates, such the Arctic or polar regions, which makes C) the odd one out.
Polar bears and seals employ this adaptation, known as blubber, to help them insulate themselves. A) Thick layer of fat under the skin.
B) White fur: Several animals have developed white fur as a form of camouflage to help them blend in with their surroundings.

D) This adaptation is typically seen in animals with large, broad feet and sharp claws, such as polar bears, which makes it simpler for them to walk about, grab icy surfaces, and effectively catch prey.

Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

26. Correct option: (C) Mix with the surroundings

Tigers and leopards have evolved to have stripes on their body, which gives them the ability to blend in with their environment and serve as an excellent kind of camouflage. Their ability to blend in and approach their target covertly is especially helpful for their hunting tactics.

In the dappled light and shadows of their native environments, such as grasslands and woodlands, the stripes blur the shape of their bodies, making it more difficult for their prey to identify them. Their ability to blend in with their surroundings and approach their victim undetected until they are prepared to strike is crucial to their effectiveness as predators.

27. Correct option: (B)

28. Correct option: (D) a – iv; b – ii; c – i; d – iii

a. Red-eyed frog has iv. Sticky pads on feet. They can climb trees because to this adaptability. Arboreal creatures, red-eyed frogs spend the most of their time in trees. Their feet include sticky pads that allow them to effortlessly move around in trees and hold onto branches.

b. Penguin has ii. Streamline body. This modification enhances their capacity for swift and effective swimming. Penguins can hunt fish and other prey using their streamlined bodies since they are skilled divers and swimmers.

c. Tiger has i. Very sensitive hearing. They have the capacity to hear noises that are imperceptible to humans. This modification makes it easier for them to discover prey and recognize predators.

d. Lion-tailed macaque has iii. Silver-white mane. They can attract mates thanks to this modification. Additionally, the main makes lions appear bigger and more frightening, which can aid in fending off predators.

So, the answer is D. a – iv; b – ii; c – i; d – iii.

29. Correct option: (A) a – iv; b – iii; c – ii; d – i

a) Elements of weather - iv) Temperature, humidity: The term "elements of weather" refers to a variety of elements, such as temperature and humidity, that make up the atmospheric conditions.

b) Cause of all changes - iii) Sun: The sun is the main influence on Earth's weather and climate. All alterations in weather patterns and atmospheric conditions are influenced by its energy.

c) Tropical region - ii) Hot and wet: High temperatures and high humidity combine to create a hot and humid environment in tropical locations.

d) Desert climate - i) Hot and dry: Low humidity and high temperatures create a dry and warm atmosphere in desert settings.

30. Correct option: (B) (i), (iii)

The conditions from the above list that are responsible for the presence of a large number of plants and animals in tropical rainforests are:

(i) Humid and hot climate: Tropical rainforest’s humid and hot climates are ideal for the development and spread of a wide range of plant species. High temperatures and humidity encourage quick plant development, which results in a varied and thick forest canopy.

(iii) Abundant rainfall: Tropical rainforests get an abundance of rainfall, which guarantees a steady supply of water, which is essential for the survival and development of both plants and animals. It supports a wide range of plant growth and offers home to a large number of animal species.

9. Garbage in and Garbage Out
Solutions

LEVEL – I

1. Correct Option (B)

The right response is (b) Eggshells, tea leaves, and fruit and vegetable peels. Decomposing organic materials into nutrient-rich humus through the process of composting can enhance the soil's quality. It's crucial to use organic material that will decompose naturally in this process. Eggshells, tea leaves, and fruit and vegetable peels are examples of items high in organic matter and nitrogen that are advantageous for composting. They give the compost pile's microorganisms the nutrients they need to thrive and convert organic matter into compost that is useful. In contrast, materials like plastic, polythene bags, paper, tin foil, and leather are included in options (a), (c), and (d), which are not suitable for composting because they are not biodegradable or because they can introduce dangerous contaminants into the compost.

2. Correct Option (B)

Due to their reduced risk of flooding, landfills are typically found in low-lying areas. A landfill's waste may become contaminated by flooding, and the surrounding area may become contaminated as well. Furthermore, low-lying areas are frequently less desirable for other purposes, like agriculture or habitation.

In order to keep waste contained and keep it from polluting the environment, landfills were created. To stop groundwater from leaking into the waste, they have plastic or clay linings. In order to keep the waste from blowing away or luring pests, it is also covered with soil.

Although landfills are a necessary component of waste disposal, they can have detrimental effects on the environment. They have the ability to release methane, a greenhouse gas, into the air. They can also contaminate groundwater and soil. However, there are ways to minimize the environmental impacts of landfills, such as using methane capture and treatment systems.

3. Correct Option (C)

The answer is (c) Eggshells, vegetable and fruit peels, and tea leaves. These materials are great options for composting because they are organic, can decompose naturally, and can add vital nutrients to the compost. Tea leaves add nitrogen to the mix, fruit and vegetable peels offer a variety of organic matter, and eggshells contribute calcium. The items listed in option (c) are perfect components for creating nutrient-rich compost that can be used to improve the quality of the soil for gardening and farming, while options (a), (b), and (d) include non-biodegradable or unsuitable materials for composting. Composting is an eco-friendly way to recycle kitchen and garden waste.

4. Correct Option (D)

The items in the options list all contribute to waste in various ways. Despite having a medical use, the syringe a doctor uses turns into hazardous waste and needs to be properly disposed of. A soft drink can and a food wrapper are examples of single-use items that, after being used, typically end up as litter or non-biodegradable waste in landfills. In order to reduce their negative effects on the environment, all three of the items mentioned in the options should be responsibly disposed of or recycled whenever possible.

5. Correct Option (C)

Recycling is a waste management technique that involves processing and using materials from discarded items to make new products, not as a means of disposing of trash. Recycling is the process of gathering, sorting, and reusing waste materials like paper, glass, plastic,

and metal to create new products with less raw material consumption and less environmental impact.

The other three choices—landfill, vermicomposting, and composting—all involve getting rid of waste. In a landfill, waste is disposed of in a specific location. Worms are used in the vermicomposting process, which breaks down organic waste. Utilizing microorganisms to break down organic waste is known as composting.

Recycling is a more environmentally friendly method of waste disposal than composting, vermicomposting, or landfilling. Recycling promotes conservation.

6. Correct Option (D)

Garbage goes through several stages of decay when it rots or decomposes, during which time organic material is broken down into simpler compounds. When the waste has fully decomposed and no longer emits any detectable odors, the process is deemed to be finished. The organic materials at this point have developed into a stable, humus-rich compost or soil conditioner that can be used to improve the soil for farming or gardening. When all the organic material has decomposed and any offensive odors have disappeared, leaving behind a useful and odorless resource for soil improvement, properly managed compost piles reach this stage.

7. Correct Option (C)

Green is typically the color of the trash can used for biodegradable or organic waste, such as food scraps, fruit and vegetable peels, and yard waste. The proper segregation of waste is facilitated by the widespread use of this color-coding system for trash cans throughout the world. Biodegradable waste is placed in green bins because it can be composted to produce valuable organic matter for soil improvement. Utilizing different colored bins makes it simpler for people to correctly sort their trash, encouraging recycling and eco-friendly waste management techniques.

8. Correct Option (A)

Sewage is the term for the domestically produced liquid waste. Wastewater from different domestic sources, such as toilets, sinks, showers, and washing machines, is referred to as sewage. Human waste, soaps, detergents, and other contaminants are all mixed together in it. Before sewage is released into the environment or recycled as treated water for various uses, it must be properly collected and treated to remove contaminants and pathogens. Since untreated sewage can present serious environmental and health risks, effective sewage treatment is essential to prevent water pollution and safeguard public health.

9. Correct Option (C)

Wastepaper is the option (c) on the list that can be recycled. Paper waste, including office paper, cardboard, and magazines, can be gathered, processed, and recycled to create new paper goods. Paper recycling lessens the environmental impact of paper production by saving energy and trees. On the other hand, organic materials like animal waste and kitchen scraps can be composted rather than recycled through conventional recycling processes, while leather shoes (a) are typically not recyclable due to their complex materials and construction.

10. Correct Option (A)

One type of waste that can be recycled is plastic. Plastic is a non-biodegradable substance that takes hundreds of years to break down naturally, in contrast to organic wastes like onion peel, eggshell, and banana peel. Recycling plastic saves energy and resources that would otherwise be used in the manufacture of new plastic, as well as reducing the amount of waste that is dumped in landfills or the ocean. Recycling plastic waste allows it to be turned into fresh goods like bottles, containers, and even clothing, which promotes a sustainable and circular economy.

On the other hand, onion peels (b), eggshells (c), and banana peels (d) are organic waste materials that are typically composted rather than recycled.

11. Correct Option (D)

All of these things should be disposed of according to standard waste disposal procedures in the garbage. Because it is frequently made of difficult-to-separate and recyclable mixed materials, the wrapper of a biscuit pack (a) is typically not recyclable. The tin of a cold drink (b) should also be thrown out in the trash because recycling it might be challenging given that it has an interior lining and might still contain beverage remnants. In order to avoid contamination and potential harm, the syringe a doctor used (c) should be properly disposed of as medical waste by adhering to safety protocols. Therefore, for safe and responsible disposal, each of these items should be placed in the proper waste containers.

12. Correct Option (A)

Earthworms play a crucial role in the composting process and the production of nutrient-rich organic matter, or manure. They consume organic waste products like food scraps and decaying plant matter, breaking them down into castings that are rich in nutrients and an important part of natural fertilizers and manure. In contrast to earthworms, which are typically involved in the decomposition or composting of organic matter, houseflies, eagles, and crows are not. Rather, they fill various ecological niches and play different roles in the environment, such as scavenging or hunting for food.

13. Correct Option (C)

Making compost through vermicomposting is option (c). With the help of vermicomposting, it is possible to efficiently and sustainably dispose of leaves that fall from trees by converting them into nutrient-rich compost. Earthworms are used in vermicomposting to convert organic matter, such as leaves, into castings that are rich in nutrients. This process reduces waste while also producing a valuable soil conditioner that can improve plant growth and soil fertility. Burning leaves is not environmentally friendly and can release harmful pollutants into the air, whereas boiling and drying leaves to make compost is an energy-intensive process that is less effective than vermicomposting. Vermicomposting is thus the most advantageous and environmentally responsible method of getting rid of tree debris.

14. Correct Option (A)

Usually, waste from cities is collected and disposed of at designated landfill sites. For the safe and controlled disposal of solid waste produced by urban areas, landfills are engineered buildings. These locations are carefully managed to reduce their negative effects on the environment and prevent soil and groundwater contamination. To manage and contain the waste, it is deposited, compacted, and covered with layers of soil or other materials in these landfill areas, where it is gathered from homes, businesses, and other sources. The other alternatives, near every colony (d), inside ponds and lakes (c), and along riversides (b), are inappropriate and environmentally harmful places to dispose of municipal waste because they can harm ecosystems and cause pollution.

15. Correct Option (A)

Waste paper is a prime candidate for recycling because it can be recycled using a variety of techniques to make new paper products. Recycling used paper reduces the need for new wood pulp, protecting forests and other natural resources while reducing energy use during the papermaking process. Conversely, leather shoes (b) may be donated or used for another purpose but are typically not recycled. Since they naturally decompose into nutrient-rich compost that can be used to improve soil for gardening or agriculture, animal waste (c) and kitchen waste (d) are organic materials that can be composted rather than recycled.

16. Correct Option (C)

Due to its composition, which prevents it from decomposing naturally over time, aluminium foil is a waste that cannot decompose. Aluminum foil is made from a metal and does not decompose in the same way as paper, which is made from plant fibres and has that ability. Therefore, aluminium foil that is thrown away in the environment will do long-term environmental harm and contribute to pollution. Aluminium foil is the non-biodegradable waste, so far, among the options presented.

In contrast, options like paper (a), faecal matter (b), and cotton (d) are biodegradable materials. Paper and cotton are derived from plant fibers and can decompose over time when exposed to environmental conditions, while faecal matter is organic waste that can also biodegrade and return nutrients to the soil.

17. Correct Option (A)

The acronym "3'R" stands for three critical environmental preservation and sustainable waste management principles. By using fewer resources and products, "reduce" encourages waste minimization, "reuse" encourages the practice of reusing items multiple times to increase their lifespan, and "recycle" emphasizes the process of converting materials and products into new items to lessen the need for fresh raw materials and energy. Together, these three ideas support a more environmentally responsible and sustainable method of resource conservation, waste management, and environmental protection.

18. Correct Option (D)

Due to the fact that bread is an organic substance that can naturally decompose, it will not pollute the soil. In the long run, bread left in the soil will decompose and supply nutrients to the nearby plants and animals. Thermocol, plastic, and aluminium foil, on the other hand, are all non-biodegradable substances that take a long time to break down. These substances have a long half-life in the soil and can harm the environment and cause pollution. Therefore, the only option among those presented that won't contaminate the soil is bread.

19. Correct Option (A)

Microorganisms like bacteria and fungi are primarily responsible for rotting, or decomposition. The breakdown of organic matter, such as dead plants, animals, and other organic materials, is greatly aided by these tiny organisms. Enzymes that are secreted by microbe’s aid in the breakdown of complex organic compounds into less complex ones so that the microbes can then absorb the simpler compounds as nutrients. This decomposition procedure is a crucial component of nature's waste disposal system and is necessary for recycling nutrients back into the ecosystem. Although insects (b) and ants (c) might make a small contribution to the deterioration of organic matter, microbes actually play the majority of the role in rotting. Amoeba (d) is a single-celled organism that does not typically participate in the rotting or decomposition process.

20. Correct Option (B)

Typically, (b) Blue bins are used to collect recyclable materials. In numerous recycling programs all over the world, blue bins are frequently used for the collection of recyclable materials. Paper, cardboard, plastic bottles, glass containers, and metal cans are all acceptable materials to place in these bins because they can all be recycled to make new products. By making it simple for people to distinguish between recyclables and non-recyclable waste, the color coding encourages recycling and ethical waste disposal practices. Regional differences in bin color may exist, but recycling bins are typically blue in color.

21. Correct Matching: a-iv, b-v, c-ii, d-iii, e-i

(a) Landfill - (iv) Land where garbage is thrown: Landfills are places set aside for disposing of solid waste, including garbage. They are substantial locations where waste is dumped and frequently covered with dirt or other materials.

(b) Vermicomposting - (v) Method of preparing compost: Vermicomposting is a composting technique in which organic matter, such as kitchen scraps, is broken down by worms, usually red worms, to create nutrient-rich compost.

(c) Red worms - (ii) Form compost: Earthworms of the red variety are used in vermicomposting. They consume organic waste and aid in the breakdown of materials to create compost.

(d) Recycling - (iii) Making things from waste: Recycling is the process of gathering and processing waste materials in order to transform them into new products. Waste materials are converted into manufacturing raw materials in this process.

(e) Plastic - (i) Give out harmful gases on burning: Burning plastic may release toxic gases into the atmosphere that could harm human health and contribute to air pollution. Due to the risks to both the environment and human health, burning plastic is generally discouraged.

LEVEL – II

1. Correct Option (C)

The answer is (c) Waste material that is disposed by humans. Organic waste that can naturally decompose over time with the aid of microorganisms like bacteria and fungi is classified as biodegradable waste. Food scraps, vegetable peels, fruit rinds, and other organic matter are frequently found in kitchen waste. These materials can be turned into nutrient-rich compost, which can be used as a natural fertilizer to improve soil, when properly disposed of, such as through composting. As a result, kitchen waste can be used for recycling and sustainable waste management strategies, which helps to lessen the amount of organic waste that is dumped in landfills and promotes the health of the soil and plant growth.

2. Correct Option (A)

Kitchen waste is considered as "biodegradable" waste. Organic waste that can naturally decompose over time with the aid of microorganisms like bacteria and fungi is classified as biodegradable waste. Food scraps, vegetable peels, fruit rinds, and other organic matter are frequently found in kitchen waste. These materials can be turned into nutrient-rich compost, which can be used as a natural fertilizer to improve soil, when properly disposed of, such as through composting. As a result, kitchen waste can be used for recycling and sustainable waste management strategies, which helps to lessen the amount of organic waste that is dumped in landfills and promotes the health of the soil and plant growth.

3. Correct Option (B)

Typically, garbage is gathered by cleaning crews and transported to landfill sites, which are low-lying areas set aside for the disposal of solid waste. Landfills are constructed buildings intended for the safe management and containment of waste. To reduce the environmental impact and control odors, waste is compacted and covered with soil or other materials before being dumped in designated areas by garbage trucks at landfills. To avoid pollution, safeguard groundwater, and guarantee that waste is appropriately contained and handled in an environmentally responsible way, landfills are carefully managed. Septic tanks and sewage tanks, on the other hand, are intended for the treatment of wastewater rather than the disposal of solid waste.

4. Correct Option (C)

A landfill can be turned into a park once it is entirely full and no longer useful for the disposal of waste. This procedure, which is a component of landfill reclamation, is regarded as an eco-friendly way to use land that had previously been used for waste disposal. A properly closed and covered landfill can be converted into a park, a green space, or other public amenity, serving as a valuable resource for the neighborhood. The procedure typically entails landscaping the area to make it safe and aesthetically pleasing for public use, covering the landfill with soil, and installing a protective liner. Former landfills being turned into parks not only helps to improve the local environment but also encourages sustainable land use and neighborhood growth.

5. Correct Option (A)

Dry leaves fall under the "biodegradable" waste category. Organic materials that naturally decompose over time with the aid of microorganisms like bacteria and fungi are referred to as biodegradable waste. Dry leaves can be composted or allowed to naturally decompose in the environment because they are organic plant material. They break down into nutrient-rich humus, which can be applied to the soil as natural fertilizer, when they decompose. As a result, dry leaves become a valuable resource for composting and a sustainable method of recycling organic material, lowering the amount of waste that is dumped in landfills.

Waste that cannot be broken down by microorganisms is referred to as non-biodegradable waste. Plastic, glass, and metal waste are a few non-biodegradable waste examples.

Composed waste is waste that has undergone microbial degradation. Compost is a nutrient-rich soil enhancement material that can be used to enhance plant health. Waste that can be divided into different categories, such as recyclables, compostables, and non-recyclables, is referred to as gradeable waste.

6. Correct Option (B)

A plastic bag is classified as a "non-biodegradable" waste. Materials that do not naturally decompose or break down into harmless substances over time when disposed of in the environment are considered non-biodegradable. Synthetic polymers with high resistance to microbial decomposition, like polyethylene, are the main constituents of plastic bags. Because of this, when plastic bags are thrown away, they can stay in the environment for a long time and cause litter, pollution, and harm to wildlife. To lessen their damaging effects on the environment, there is an increasing focus on reducing the use of plastic bags and promoting their recycling or proper disposal.

7. Correct Option (B)

The rotting and conversion of some material into manure is called "composting." Composting is a natural process in which microorganisms like bacteria, fungi, and enzymes transform organic waste products like kitchen scraps, yard waste, and other biodegradable materials into nutrient-rich humus-like material. The conditions for decomposition are created by layering, turning, and aerating the organic materials in a controlled environment, usually a compost pile or bin. These materials eventually break down into compost, a beneficial fertilizer and soil conditioner that can be used to strengthen the soil and promote plant growth. Therefore, composting is an environmentally friendly and sustainable method for turning organic waste into nutrient-rich manure, and it aids in lowering the amount of waste that ends up in landfills while fostering soil sustainability.

8. Correct Option (A)

The organisms in vermicompost, specifically a species known as Eisenia fetida or red wiggler worms, are essential for converting organic waste into usable manure. Due to their voracious appetite for organic matter, including food scraps and other decomposing plant materials, these worms are ideal for composting. As the redworms eat this organic waste,

their digestive systems break it down, and their castings—their excrement—become rich in nutrients and advantageous microorganisms. These "black gold" worm castings are excellent organic fertilizers and soil conditioners, adding vital nutrients and enhancing soil structure for more robust plant growth. Redworms are therefore crucial to the vermicomposting process, assisting in the transformation of organic

9. Correct Option (D)

Plastics, metals, and glass fragments are among the recyclables that are frequently collected in the blue bin. This container is a part of a recycling programme that separates and recycles these materials in an effort to cut waste and save resources. Glass pieces refer to glass bottles and containers, while plastics cover a wide range of plastic goods and packaging, while metals include things like steel containers and Aluminium cans. By putting all of these recyclable materials in the blue bin, people can help the recycling process, in which waste is gathered, sorted, and processed to create new products, thereby promoting resource conservation and environmental sustainability. Option (d), "All the given options," is the right response because it includes all of the collected recyclable materials.

10. Correct Option (B)

Dry leaves are an extremely biodegradable form of organic plant material. Dry leaves easily decompose when added to soil thanks to the action of microorganisms like bacteria and fungi. The soil is enriched for plant growth by these microorganisms as they convert the leaves into organic matter and release nutrients. In contrast, non-biodegradable alternatives like plastic, glass fragments, and nylon strings are difficult for the soil to break down. They have a long shelf life and can pollute the environment and harm ecosystems. Dry leaves are an excellent resource for composting and soil enrichment because they can easily and naturally decompose in the soil as waste.

11. Correct Option (B)

Red worms, also referred to as earthworms, lack teeth, unlike people. Instead, they can grind and digest their food with the aid of a special organ called a gizzard. The gizzard is a muscular organ in the worm's digestive tract that is a part of its digestive system. The gizzard contracts after ingesting small mineral particles and grit, which it uses to grind the food when the worms consume organic matter like leaves or other decomposing plant material. This physical breaking down of the food into smaller pieces facilitates the further breakdown and digestion of the organic matter by the worms' digestive enzymes. So, the gizzard plays a crucial role in the earthworm's digestion process by aiding in the mechanical breakdown of food.

Animals use their teeth to tear and chew their food. Red worms are unable to tear or chew their food because they lack teeth. Additionally, they lack the radula, an organ resembling a tongue that some animals use to scrape food into their mouths. Red worms, on the other hand, eat their food whole and rely on their gizzard to grind it.

12. Correct Option (C)

The correct answer is (c) Reuse. Reusing something means using it without significantly altering it or processing it extensively. Finding new applications for an item after its initial use—either in its current form or with only minor modifications—instead of throwing it away after use. Making the most of already-existing materials promotes resource conservation, lowers waste generation, and increases product lifespan. In contrast, choices like "Reduce" put the emphasis on cutting back on consumption, "Recycle" entails using old materials to make new products, and "Redraw" has nothing to do with using or modifying existing things. Reusing something means using it without significantly processing it or changing it, which is consistent with the definition of reuse.

13. Correct Option (B)

Recycling is the process of using waste materials that have been used up or discarded to make new products or materials. To do this, recyclable materials like paper, glass, plastics, and metals must be gathered, sorted, cleaned, and converted into raw materials that can be used to create new goods. Recycling is a crucial component of sustainable waste management because it aids in resource conservation, lowers energy usage, and lessens the environmental impact of producing new materials. Therefore, "Recycle" is the term that specifically refers to processing something to create a new product from previously used or discarded materials among the options offered.

14. Correct Option (D)

Plastic should not be thrown down drains, into water tanks, or out of water outlets as it may obstruct the flow. Plastic waste that enters drains has a tendency to build up and clog the drainage system, which can cause waterlogging and flooding during periods of heavy rain. Plastic waste can contaminate the water supply and make it more challenging to filter and purify the water if it is put into water tanks. Similar to plastic waste, obstructions in water outlets can stop water flow, causing backups and possibly damaging infrastructure. Because of this, disposing of plastic waste in these places is not only bad for the environment but can also interfere with key services and result in a number of issues.

15. Correct Option (A)

When compared to making paper from scratch, recycling paper is an environmentally friendly strategy that not only lessens the need to clear forests of trees but also saves energy and resources. When paper is recycled, it goes through a process that makes new paper products out of it, extending its useful life and reducing the amount of waste that is dumped in landfills. While recycling is particularly effective in ensuring that paper materials are repurposed and made into new items, contributing to a more sustainable and environmentally friendly approach to managing paper waste, reusing and composting paper are also beneficial methods for reducing waste.

16. Correct Option (C)

The organic waste that can naturally decompose over time is called biodegradable waste, and it is kept in the green dust bin. This includes things like leftover food, garbage from the kitchen, grass clippings from the garden, and other organic materials that decompose into compost or can be recycled naturally. In order to encourage environmentally friendly disposal techniques, like composting or recycling, rather than sending biodegradable waste to landfills where it contributes to the production of methane gas and other environmental issues, green bins are typically designated for this type of waste. As a result, option (c), "biodegradable waste," is the appropriate response because it fits with the objective of green dust bins.

17. Correct Option (C)

Burning waste causes air pollution, as indicated by option (c). Burning waste materials releases dangerous pollutants into the atmosphere, especially if they contain plastics, paper, or other synthetic materials. Particulate matter, carbon monoxide, Sulphur dioxide, nitrogen oxides, volatile organic compounds, and dangerous chemicals are just a few examples of these pollutants. Humans who breathe in these pollutants may experience serious respiratory and cardiovascular problems as well as other negative health effects. The fact that burning waste releases greenhouse gases also emphasizes the harm that this type of pollution does to the environment by causing climate change. As a result, burning waste is a major cause of air pollution.

18. Correct Option (B)

We do not use plastic bags to store eatables, as indicated by option (b). Because plastic bags are made of synthetic materials that can leach dangerous chemicals into food, they are not appropriate for storing food items. When these chemicals come into contact with edible products, such as phthalates and bisphenol A (BPA), health risks may result. As opposed to plastic bags, which are more likely to contaminate food than jute, nylon, and polyester bags, these materials are generally thought to be safer options for storing food. Therefore, due to the potential health risks associated with their use, plastic bags are not advised for food storage.

19. Correct Option (D)

All the options listed will add to garbage. Kitchen waste, such as food scraps and leftovers, contributes to organic waste in garbage. The wrapper of a biscuit is typically made of non-biodegradable materials like plastic or Aluminium foil, which also ends up in the garbage. Similarly, the tin of a cool drink, made of metal or Aluminium, is not biodegradable and becomes part of the garbage when discarded. Therefore, each of these options, (a) Kitchen waste, (b) Wrapper of a biscuit, and (c) Tin of a cool drink, adds to the overall volume of garbage when they are disposed of improperly.

20. Correct Option (C)

Wastepaper is a type of paper that has been used previously but can still be processed to create new paper products. Recycling used paper helps preserve forests and uses less energy and resources than manufacturing new paper from scratch. In contrast, animal and kitchen waste can be composted to produce soil rich in nutrients, but these organic materials are not typically recycled in the same way as paper products. Wastepaper is the only option from the ones provided that can be recycled to make new products.

21. Correct Option (C)

The term "3'R" is a sustainability concept that encourages responsible resource management and environmental conservation.

Reduce: This entails cutting back on resource consumption and waste production. It encourages people and companies to use resources more wisely, which can include buying energy-efficient appliances, saving water, and using fewer single-use items.

Reuse: Reuse encourages reusing goods or resources rather than discarding them after a single use. This can entail doing things like donating clothing and goods rather than throwing them away and using reusable containers and shopping bags.

Recycle: Recycling is the process of gathering and processing materials into new products that would otherwise be thrown away as waste. It helps keep resources in check, uses less energy, and keeps waste out of landfills.

22. Correct Option (B)

Microorganisms like bacteria, fungi, and other microbes play a major role in the process of rotting, also known as decomposition. These tiny creatures are essential for the transformation of organic matter, such as excreta from dead animals and plants, into simpler substances. They accomplish this by secreting enzymes that disassemble intricate organic molecules into simpler parts that the ecosystem can absorb and recycle. Microbes are in charge of the environment's natural recycling of nutrients and organic materials, which makes it possible for crucial elements like carbon and nitrogen to return to the soil and support plant growth and ecological balance. Therefore, decay is a biological process fueled by microbes.

23. Correct Option (C)

Organic materials like dry leaves naturally decompose, supplying the soil with vital nutrients as they do so. Dry leaves are preferable to drainage water (option b), which can

introduce pollutants and contaminants into the soil, and plastic (option a), which is non-biodegradable and can persist in the soil for hundreds to thousands of years. In addition to decomposing, they also improve soil structure, moisture retention, and promote advantageous microbial activity. Due to their advantageous qualities, dry leaves are actually widely used by farmers and gardeners as a form of mulch or as a component of compost. Dry leaves therefore benefit the environment rather than polluting the soil.

24. Correct Option (C)

The correct answer is (c) Nutrients. Compost that is rich in nutrients is created when organic waste, such as plant waste and food scraps, completely rots during the composting process. Nitrogen, phosphorus, potassium, and a variety of micronutrients are just a few of the many nutrients that compost is a great source of. As the compost decomposes more, these nutrients are gradually released, giving plants a natural and balanced supply of nutrition. Compost adds these crucial nutrients to the soil, enhancing its structure and encouraging the growth of healthy plants and higher yields. Option (c) is the best option because composting allows garbage to completely rot while also giving plants valuable nutrients.

25. Correct Option (B)

When added to soil, compost is a valuable organic material that enriches it with organic matter and crucial nutrients. Compost is well known for its capacity to enhance soil structure, boost nutrient availability, and encourage advantageous microbial activity. These characteristics make the soil more fertile, which makes it more conducive to plant growth. Increased crop yields and healthier gardens result from the conditions and nutrients that fertile soil offers plants for healthy development. So, improving soil's fertility and overall productivity can be accomplished by adding compost to the soil.